VECTO 5.x Release Notes



VECTO v5.1.4-RC (03-09-2026)

Features

- Extrapolate EHC data from LHE and RD (vecto/vecto!601)
- Deploy local beta releases (vecto/vecto!608)
- Optimized torque splitter (partial implementation) and PCC torque splitter (vecto/vecto!612)
- Added ‘ExtraTechnologies’ to monitoring data (vecto/vecto!626)
- Improve error reporting (vecto/vecto!635)
- vehicle int-test script in LEGENT server (vecto/vecto!642)
- Improve error reporting part2 (vecto/vecto!641)
- Consider CRF delta in report (vecto/vecto!645)

Bug Fixes

- Battery-only P2 (vecto/vecto!580)
- Output writing for FCHVs (vecto/vecto!581)
- Consider ratio per gear for calculation of EM shift polygons in case of P2.5… (vecto/vecto!582)
- Extend GearShift cond when doCoastOvl for AMT (vecto/vecto!584)
- Add AMT 1st Gear DownShift safety (vecto/vecto!583)
- angledrive component file hashing (vecto/vecto!589)
- validation of IEPC data coming from a VIF (vecto/vecto!590)
- Unit Tests after multiple powertrains merge (vecto/vecto!592)
- PreRunOptions.WriteModAndSumData outputs (vecto/vecto!591)
- bugs created by multiple-powertrain refactoring. (vecto/vecto!597)
- Added missing mocked property for vehicle container (vecto/vecto!598)

- Add inertia torque reserve in Upshifts (vecto/vecto!595)
- Removed duplicate clutch from P1 vehicles (vecto/vecto!599)
- Validation & serialization vecto rundata in case of multiple powertrains, gearbox bindings multiple powertrains (vecto/vecto!600)
- Vocational output for group 16 vehicles (vecto/vecto!586)
- Added missing IPowertrainContainer parameter to creation methods of battery-only hybrids. (vecto/vecto!602)
- Added missing parameters to component factory methods of tests (vecto/vecto!603)
- Added property to VectoRunData for generator of multiple-powertrain SHEV (vecto/vecto!604)
- Removing unused classes (vecto/vecto!605)
- Correctly identify VIF report for mulitple-powertrain vehicles in HashingTool (vecto/vecto!606)
- error in validation of auxiliaries for battery-only hybrids (vecto/vecto!607)
- Engine fan's power demand ignored when powertrain has no engine (vecto/vecto!610)
- Property HybridElectricHDV returns true for vehicle data originating from hybrid VIFs (vecto/vecto!613)
- Add further VectoTrailer projects (tests), build script publish (vecto/vecto!614)
- Added missing components to simple powertrains for multiple-powertrain vehicles. (vecto/vecto!615)
- Added class1s vehicles for VTP selected mission: Regional Delivery (vecto/vecto!617)
- Debugged optimized torquesplitter (vecto/vecto!618)
- Improved error message for missing auxiliaries in completed vehicles (vecto/vecto!621)
- Added missing gearbox configuration for dependency injection (vecto/vecto!622)
- Reverted type of dummy gearbox back to APTN (vecto/vecto!624)
- Write axle number correctly in Monitoring Report & validate report in tests (vecto/vecto!625)
- Proper creation of IEPC completed vehicle in multistep tool (vecto/vecto!627)
- Allow v2.6 AirDrag component to be read for input in multistep tool (vecto/vecto!628)
- 1187cor - apply inertia trq reserve only if tqInterrupt (vecto/vecto!630)
- Adaptions for Vecto Trailer (vecto/vecto!629)
- Get proper segment from completed buses to retrieve CdxA for reports (vecto/vecto!632)
- P2.5 battery-only complete bus simulation aborts (vecto/vecto!631)
- Add build option to build script to create a release candidate build (vecto/vecto!633)
- problems of error handling (vecto/vecto!636)
- Caught exception when raising log event (vecto/vecto!637)
- Avoid null reference in 'Error Details' UI (vecto/vecto!638)
- Properly read certification number of fuel-cell component (vecto/vecto!639)
- Improved 'Error Details' UI component and GUI output messages. (vecto/vecto!640)
- CIF summary entry (vecto/vecto!644)
- Disable vehicles with non-identical axle-powertrains (vecto/vecto!646)

- Multiple-powertrains - torque share for coast or roll (vecto/vecto!648)
- Avoid null reference when calculating gearbox loss during initialization (vecto/vecto!649)
- Removed obsolete target .NET 8 from test and auxiliary projects (vecto/vecto!651)
- errors and updated information in README file (vecto/vecto!652)
- Added missing error codes (vecto/vecto!656)
- Server problems (vecto/vecto!653)
- label placement for cycle run info values (vecto/vecto!658)
- Prevent exception in gearbox loss calculation (vecto/vecto!659)
- Changed values for ‘Manufacturer’ and ‘Certification Number’ elements in… (vecto/vecto!661)
- Can open GUI documentation links when running VECTO from netXX folder (vecto/vecto!666)
- Overwriting vmods for FCHV OVC vehicles (vecto/vecto!671)
- Correct DS detect Condition for Brake after Acc UL (vecto/vecto!670)
- Coasting action after reengage (vecto/vecto!669)
- Change driveOffTorque APT Threshold for reengage (vecto/vecto!668)
- EM generic effmap for EM with asymmetric perfo. (vecto/vecto!667)

Refactor

- Refactoring of multiple powertrains (vecto/vecto!587)
- Removed obsolete methods from IDataBus and increased component access through IPowerContainer properties (vecto/vecto!611)
- Removed some built-time warnings (vecto/vecto!619)
- Removed build-time warnings (vecto/vecto!623)
- Kept only one property denoting position in the electric motor class (vecto/vecto!643)
- Improve error reporting part3 (vecto/vecto!650)
- Improved error messages and added suggestions (vecto/vecto!654)
- Making methods virtual in moddatacontainer and allow overrides in trailer project (vecto/vecto!665)

Testing

- Added HybridStrategyTests (vecto/vecto!593)

Improvement

- Set internals visible to trailer project (vecto/vecto!609)

VECTO v5.1.3 Official Release (03-08-2026)

This version replaces v5.1.1 which had to be retracted. It is the same in terms of functionality and results.

Features

- multiple powertrains completed and single bus. (vecto/vecto!516)
- Single-gear IEPC in multiple powertrains (vecto/vecto!524)
- Added input (axle-number specific) validation for multiple-powertrain vehicles. (vecto/vecto!537)
- PCC for multiple powertrains. (vecto/vecto!565)

Bug Fixes

- Vecto console references (vecto/vecto!507)
- Completed bus input generation for v2.7 (vecto/vecto!508)
- DefaultDriverStrategy - add accel action after gearshift + roll + brake if overload (vecto/vecto!510)
- Add retarder losses to TC max torque request (vecto/vecto!511)
- Rename constructor parameter to match factory method (vecto/vecto!512)
- Propagate ‘allowDeprecated’ parameter for XMLReader classes through injection hierarchy (vecto/vecto!513)
- Multiple powertrain FCHV bus execution (vecto/vecto!515)
- vsum fields for Retarder, Axlegear for multiple powertrains. (vecto/vecto!517)
- Increase max iteration in InterpolateSearch (vecto/vecto!514)
- use of battery SoC limits (input vs default) (vecto/vecto!519)
- Create fuel cell in testpowertrain (required in shift strategy) (vecto/vecto!520)
- Added missing function to DummyRunNinjectModule (vecto/vecto!521)
- Calculation of axlegear efficiency for P4 vehicles: in case of an P4, use EM-in to calculate axlegear efficiency instead of Brake-in (vecto/vecto!522)
- Added multiple-powertrain sample jobs (vecto/vecto!525)
- Avoid null reference in DeclarationData.cs (vecto/vecto!526)
- bugs regarding multiple powertrains. (vecto/vecto!527)
- Modified Torque Converter columns to handle multiple powertrains. (vecto/vecto!528)
- OVC-HEV mode for FCHV vehicles, null reference in fuel cells. (vecto/vecto!529)
- Don’t allow unsupported variations of multiple powertrain vehicles. (vecto/vecto!530)
- Enable unsupported vehicles in development mode (vecto/vecto!531)
- Modified xArchitecture element in reports to support multiple powertrains. (vecto/vecto!533)
- Calculate TotalPropulsionPower for multiple-powertrain vehicles (vecto/vecto!534)
- Improved error message when not finding primary vehicle missions (vecto/vecto!535)
- Removed engine power from calculation of total propulsion for SHEV (vecto/vecto!536)
- Added missing parameter to method that creates gearbox. (vecto/vecto!538)
- Modified method signature to satisfy interface (vecto/vecto!539)
- Re-enabled unit tests, that required battery input data. (vecto/vecto!540)

- Added OVC info to tests' vehicle input (vecto/vecto!541)
- Ignore monitoring data when calculating job hash for reports (vecto/vecto!544)
- hashing tests (vecto/vecto!545)
- Save vehicle in job editor (vecto/vecto!546)
- Remove xml comments before calculating hash (vecto/vecto!547)
- Hashing Tool ignores comments when verifying XML content, and Monitoring… (vecto/vecto!551)
- Null reference & sample job file (vecto/vecto!552)
- Handle cycles that fail during setup of the follow-up run. (vecto/vecto!553)
- Added pre-processing for APT-N gearboxes (vecto/vecto!554)
- Overloaded equality operators in GearshiftPosition (vecto/vecto!556)
- TC dryRun Response in Motoring conditions (vecto/vecto!561)
- Brake Overload management (vecto/vecto!560)
- Handled PCC pre-processing for IEPC single-speed gearbox (vecto/vecto!562)
- Corrected EM ElectricPowerToBattery for multiple powertrains. (vecto/vecto!563)
- Added missing dummy axlegear to E4 axle-powertrain (vecto/vecto!564)
- Avoid null reference in TestPowertrain Gearboxes (vecto/vecto!566)
- verification of reports with bad job hashes (incl. Monitoring Data) (vecto/vecto!568)
- Handle primary cycles with insufficient fuel cell power (vecto/vecto!569)
- IHPC - add ignoreReason for clutch slipping (vecto/vecto!567)
- Brake Overload management when engaged (vecto/vecto!570)
- date parsing in fuel cell component hashing (vecto/vecto!572)
- Integrate RESS ovl/underload in EM response (vecto/vecto!571)
- failing unit tests for shift strategies (vecto/vecto!573)
- Integrate RESS ovl/underload in EM response - corrected (vecto/vecto!574)
- Handle InterUrban cycle setup fail in FCHV primary bus (vecto/vecto!575)
- fuel cell input in multiple-powertrain vehicle file (vecto/vecto!576)
- Modified version-checking method to accept versions without build part (vecto/vecto!577)

Refactor

- Refactored multiple powertrains builder methods (vecto/vecto!509)
- Reduced build-time warnings (vecto/vecto!518)
- Project vecto sw3/task 2/unit tests amdm3 (vecto/vecto!460)
- Modified IDataBus properties/methods to handle single & multiple powertrains. (vecto/vecto!542)
- Renamed VectoRunData properties to indicate single powertrain (vecto/vecto!543)
- Minor cleanup/refactoring to reduce build-time warnings. (vecto/vecto!548)
- Cleaned up root folder (vecto/vecto!549)

- Cleanup/refactoring to reduce build-time warnings - part II. (vecto/vecto!550)
- Reduce compile-time warnings part III (vecto/vecto!555)
- Eliminated compile-time warnings in VectoCoreTests. (vecto/vecto!557)
- Reduce compile-time warnings in new Test projects. (vecto/vecto!558)
- Minor refactoring (vecto/vecto!559)

VECTO v5.0.7 Official Release (14-10-2025)

Bug Fixes

- Release notes link in GUI dialog (vecto/vecto!480)
- Battery only mode for IHPC (vecto/vecto!481)
- Add System.Data.SqlClient to VECTO.vbproj (vecto/vecto!482)
- Treat ‘not applicable’ as ‘none’ in HeatPumpTypeDriverCompartmentType (vecto/vecto!483)
- Correct parsing of fuel cell in interim file (vecto/vecto!484)
- Proper mapping for ADC loss-map in XML (vecto/vecto!485)
- Improved error message for missing battery SoC bounds (vecto/vecto!486)
- VETO main version in dialogs (vecto/vecto!488)
- Corrected XSD dynamic charging types (vecto/vecto!491)
- Introduce new mod data postprocessing for BO-HEV (vecto/vecto!489)
- Copy monitoring data from input (for completed vehicles) (vecto/vecto!494)
- Generic vehicles engineering mode VTP files (vecto/vecto!492)
- Surround post-mortem analysis with try/catch block (vecto/vecto!495)
- Iepc gearshift (vecto/vecto!487)
- Iepc gearshift torque reserve (post MR !487) (vecto/vecto!496)
- Corrected name for Tyre in VIF (vecto/vecto!497)
- Corrected json sample file (vecto/vecto!498)
- Added missing bindings for mod-data post processing (vecto/vecto!499)
- Allow VTP with v2.4 vehicles. (vecto/vecto!503)
- Updated some sample vehicles to v2.7 (vecto/vecto!504)

Refactor

- Remove redundant code (vecto/vecto!493)

VECTO v5.0.6-RC (22-09-2025)

Features

- Update jobs in Generic Vehicles to version v2.7 (vecto/vecto!453)
- Disable v2.4 jobs (vecto/vecto!458)
- CodeEU #1140: Update multistep GUI to work with new XSDs (vecto/vecto!469)
- Multiple axles partial implementation (vecto/vecto!471)

Bug Fixes

- Fix FCHV unit tests (vecto/vecto!448)
- XMLConversionTool bug fixes, more test cases, refactoring (vecto/vecto!452)

- Bug fixes for FCHV bus (vecto/vecto!454)
- CodeEU #1147: Made FuelCell Minpower, Maxpower optional (vecto/vecto!456)
- Engine-only simulation (vecto/vecto!457)
- EM data in PHEV rundata creation (vecto/vecto!459)
- CodeEU #1164: Lifetime ranges in reports for PEV, HEV-OVC. (vecto/vecto!461)
- Work-around in ranges to make tests succeed (vecto/vecto!462)
- CodeEU #1163: Respect job's battery SoC limits (vecto/vecto!463)
- CodeEU #870, #871, #924: Extend Accelerate condition after xEV Overload (vecto/vecto!466)
- VTP generic vehicles (vecto/vecto!465)
- Extend Accelerate condition after xEV Overload
- Changed v2.6 XSD to allow DeltaCdxA_declared and DeltaTransferredCdxA value: zero (vecto/vecto!472)
- Avoid cyclic refs from !473 (vecto/vecto!475)
- CodeEU #1167: Added WheelEnd info to MRF (vecto/vecto!476)
- Angledrive mod data, and PWheel axlegear efficiency (vecto/vecto!477)
- Disable engineering mode for multiple powertrains (vecto/vecto!478)

Refactor

- FCHV iterative run strategy (vecto/vecto!449)
- Update VECTO to NET 8 (vecto/vecto!467)
- Old .NET references (vecto/vecto!468)
- MultistepTool deprecated views (vecto/vecto!470)
- Remove unnecessary usings and nugets (vecto/vecto!473)

VECTO v5.0.4-DEV (25-08-2025)

Features

- 3rd amendment reports for buses (vecto/vecto!421)
- V1.0 reports for multiple powertrain lorries (vecto/vecto!395)
- Add Diesel B100 CI fuel (vecto/vecto!399)
- MRF and Monitoring report for multiple-powertrain primary buses (vecto/vecto!410)
- battery only mode for P2 (vecto/vecto!425)
- Run simulation for H2-ICE bus (primary + completed) (vecto/vecto!432)
- FCHV bus simulation, primary & completed (vecto/vecto!433)
- Single-bus mode for FCHV (vecto/vecto!435)
- Enable all v2.7 vehicles (vecto/vecto!436)
- VTP input and formulas for buses and trucks (vecto/vecto!424)

Bug Fixes

- Exception when getting MaxWindowsSize (vecto/vecto!422)
- For FCHV, APT-S/P gearboxes simulated as APT-N. (vecto/vecto!423)
- Ovc s-hev cs cd (vecto/vecto!420)
- Convert property Type to Architecture in axle powertrains (vecto/vecto!393)
- 3 job types for multiple powertrains (vecto/vecto!394)
- Removed NgTankSystem from Multiple_SHEV primary bus (vecto/vecto!398)
- Remove Retarder component from X4 architectures (vecto/vecto!400)
- Use multiple factory methods for Retarder and Angledrive data providers (vecto/vecto!406)
- Added FCHV missing gearbox bindings (vecto/vecto!427)

- Airdrag element in VIF report. (vecto/vecto!430)
- Do not require SoC limits for HV non-OVC (vecto/vecto!431)
- B100 density to 890 kg/m3 (vecto/vecto!411)
- Run old VIFs with v2.7 Completed vehicles (vecto/vecto!434)
- 1065 vehicle co2 group (vecto/vecto!428)
- DoCoast - add drive condition for overload (vecto/vecto!426)
- Generic retarder and failing tests (vecto/vecto!437)
- IEPC data adaptation (vecto/vecto!438)
- IHPC VECTO run data (vecto/vecto!439)
- FCHV IEPC rundata gearbox creation (vecto/vecto!440)
- Generic vehicles that failed to run (vecto/vecto!441)
- WHRCharger creation (vecto/vecto!442)
- Update wheelEnd sample (vecto/vecto!443)
- FCHV files in engineering mode (vecto/vecto!444)
- Initialize MaxChargingPower from static data if not available in input (vecto/vecto!445)
- FCHV battery and CD and CS runs (vecto/vecto!446)

Refactor

- Merge refactoring branch (vecto/vecto!419)
- Merging refactorings from SW3 project to (vecto/vecto!396)

VECTO v5.0.3 Official Release (08-07-2025)

Bug Fixes

- Avoid em conditioning calculation for non-FCHV (vecto/vecto!416)
- Safe conditioning data lookup (vecto/vecto!417)

VECTO v5.0.1 Official Release (03-07-2025)

Bug Fixes

- FCHV engineering run (vecto/vecto!391)
- H2 check in bus job (vecto/vecto!392)
- Auxiliaries REESS connection (vecto/vecto!397)
- Read PowerOutputConsumptionMap as kW (vecto/vecto!401)
- Select pruned missions for FCHV primary bus (vecto/vecto!403)
- Use angledrive in lorries' gearshift data creation (vecto/vecto!404)
- Remove wrong bus angledrive restrictions (vecto/vecto!405)
- Decl GUI error message when FCHV in eng mode (vecto/vecto!407)
- Set NgTankSystem default for primary buses (vecto/vecto!408)
- FCHV F-IEPC simulation runs! (vecto/vecto!409)
- Operational range for group 10 vehicle weights (vecto/vecto!402)
- Ovc fc weighting to correspond to CO2 computation (vecto/vecto!413)

- 3s Buffer compute max EM PLoss for FL OPs (vecto/vecto!412)
- Conditioning power demand for FCHVs (vecto/vecto!414)

VECTO v5.0.0-RC (05-06-2025)

Features

- New (v2.7) XSD for vehicles (vecto/vecto!334)
- Partial implementation for new vehicle battery (vecto/vecto!337)
- Readers for v2.7 vehicle XSD, and support for fuel cell vehicles. (vecto/vecto!341)
- Read monitoring data from job (vecto/vecto!345)
- 3rd amendment mrf cif xml schemas (vecto/vecto!340)
- MRF v1.0 vehicle (lorries and FCHV primary buses) writers (vecto/vecto!354)
- CIF v1.0 vehicle part (v2.4 vehicles and v2.7 lorries) (vecto/vecto!355)
- Use monitoring data from job to write report (vecto/vecto!360)
- In motion charging postprocessing (vecto/vecto!344)
- Disable (for RC & official) v27 vehicles except H2-ICE & FCHV lorries (vecto/vecto!375)
- EM-IEPC Thermal Derating - Tq_max and Buffer Mods - Post VECTO-4.3.4 Feed-Back (vecto/vecto!378)
- Forbid AT upshift for reduced dt before brake (vecto/vecto!381)
- Readers for v2.7 buses, improved reader tests. (vecto/vecto!382)
- HEV - Get Best dSOC in vsum (vecto/vecto!383)

Bug Fixes

- Non-https link in manual (vecto/vecto!339)
- V2.7 reader & XSD (vecto/vecto!342)
- Updated Monitoring Report XSD (FCHV, Multiple powertrains) (vecto/vecto!335)
- Updated XSLT file and hashing code for new vehicles and components. (vecto/vecto!338)
- Lock StoredResults list before accessing it to avoid race condition (vecto/vecto!343)
- V27 vehicle issues (vecto/vecto!346)
- 882 merge artifacts (vecto/vecto!347)
- FCHV angledrive input (vecto/vecto!348)
- Modify schema so that results can be written compatible with results for 2nd amendment: (vecto/vecto!349)
- Restore deleted code in monitoring report (vecto/vecto!350)
- Typo in MRF Inject module (vecto/vecto!351)
- Proper namespace for VIF IEPC sub-element (vecto/vecto!352)
- Mockup tests run successfully (vecto/vecto!353)

- Correcting errors in XML schema (and sample files): no engine output in… (vecto/vecto!356)
- Bugfixes/updates for the Monitoring report and testing via the MockupTests. (vecto/vecto!357)
- Replace U+2013 by regular dashes (vecto/vecto!359)
- Added missing IMC testdata (vecto/vecto!361)
- Retarder compulsory in all MRF vehicle components. (vecto/vecto!362)
- Add further condition to decide which results to write in case the input data is a Multistep bus (vecto/vecto!363)
- Check Articulated in json vehicle (vecto/vecto!365)
- Segment in Bus AirDrag data creation (vecto/vecto!364)
- FCHV pre-run execution (vecto/vecto!366)
- Write ZeroCO2EmissionsRange and HydrogenRange to H2-ICE reports. (vecto/vecto!367)
- Remove wrong angledrive restrictions (vecto/vecto!369)
- Set vectorundata in completed bus results, (vecto/vecto!368)
- Simulate OVC for FCHVs (vecto/vecto!370)
- FCHV H2 range in reports (vecto/vecto!371)
- Architecture in some MRF v1.0 tests (vecto/vecto!372)
- H2 properties check in exempted vehicle input (vecto/vecto!373)
- FCHV input classes inheritance (vecto/vecto!374)
- Standard values enum entry for v2.6 (vecto/vecto!376)
- Add Driving Actions for IEPC gearshift (vecto/vecto!377)
- ReEngage1C tolerance in AT (vecto/vecto!379)
- Take battery limit into account for EM overload - REESS Empty (vecto/vecto!380)
- Of v2.7, allow only IMC, H2-ICE conventional, FCHV Lorries (vecto/vecto!384)
- Disable reading data from external csv (vecto/vecto!385)
- OVC results (vecto/vecto!387)
- Ignore FCHV pre-run in best deltaSoC calculation (vecto/vecto!388)

Drop

- MaxChargingPower requirement for OVC in v2.7 (vecto/vecto!358)